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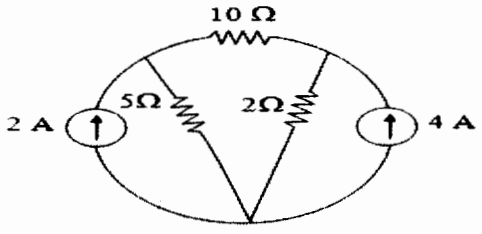
B.TECH
(SEM-III) THEORY EXAMINATION 2019-20
NETWORK ANALYSIS & SYNTHESIS

Time: 3 Hours *Total Marks: 100*

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

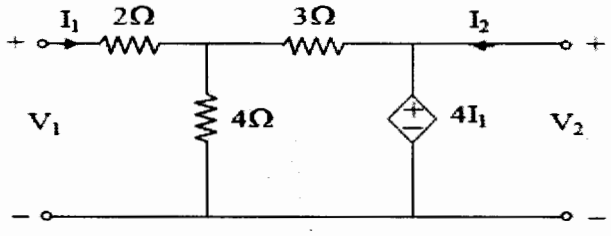
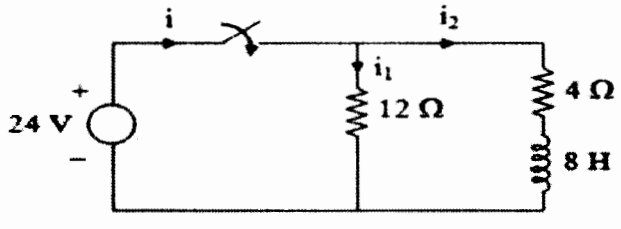
SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Qno.	Question	Marks
a.	Explain the concept of Complex Frequency.	2
b.	Define "Transfer function" of a network.	2
c.	State two properties of the R-C driving point Impedance function.	2
d.	Find the Laplace transform of $x(t) = e^{-at} \sin \omega_o t$	2
e.	Find Current in 10ohm resistor as shown in fig: 	2
f.	Draw the Dual Circuit of Parallel RLC circuit with Current Source.	2
g.	What are the Dependent & Independent terms in the Z- parameter?	2
h.	State Compensation Theorem.	2
i.	Give examples of Active & Passive elements in a Network.	2
j.	Draw the Frequency Resonance Curve of Parallel Resonance R-L-C Circuit.	2

SECTION B

2. Attempt any *three* of the following:

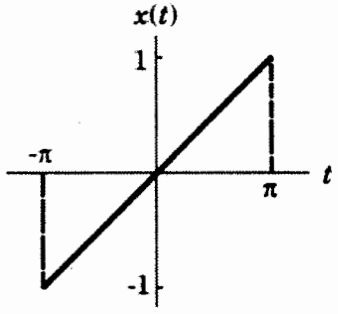
Qno.	Question	Marks
a.	Find Y and Z parameters of the networks as shown in fig 	10
b.	Explain Low pass Filter, High Pass Filter, Band Pass Filter, Band Reject Filter.	10
c.	Find the current i_2 for $t > 0$ in the circuit shown below as shown in fig 	10

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d.	Explain Maximum Power Transfer Theorem related to AC Circuits.	10
e.	Calculate the inverse Laplace Transform h(t) of given transfer function $$H(s)=\frac{s^2+5s-9}{(s+1)(s^2-2s+10)}$$	10

SECTION C

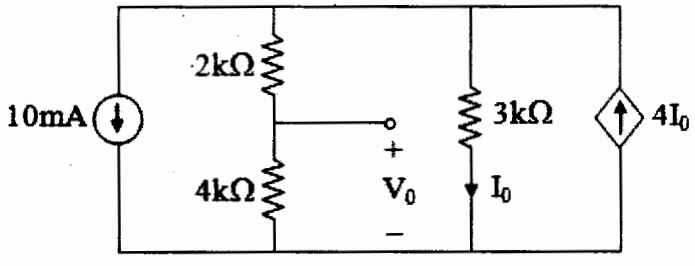
3. Attempt any *one* part of the following: (10 ×1=10)

Qno.	Question	Marks
a.	A Series R-L circuit has constant voltage V applied at t=0. At what time does $V_R=V_L$ happens.	10
b.	A periodic waveform whose one period is shown in fig. Determine the trigonometric Fourier series coefficients. 	10

4. Attempt any *one* part of the following: (10 ×1=10)

Qno.	Question	Marks
a.	Calculate the Inverse Laplace Transform using Convolution Integral. $$F(s)=\frac{1}{(s+a)(s+b)}$$	10
b.	For the Continuous time periodic signal $$x(t)=1+\cos\frac{2\pi}{3}t+4\cos\frac{5\pi}{3}t$$ Determine the Fundamental frequency w_0 & exponential Fourier series co-efficients.	10

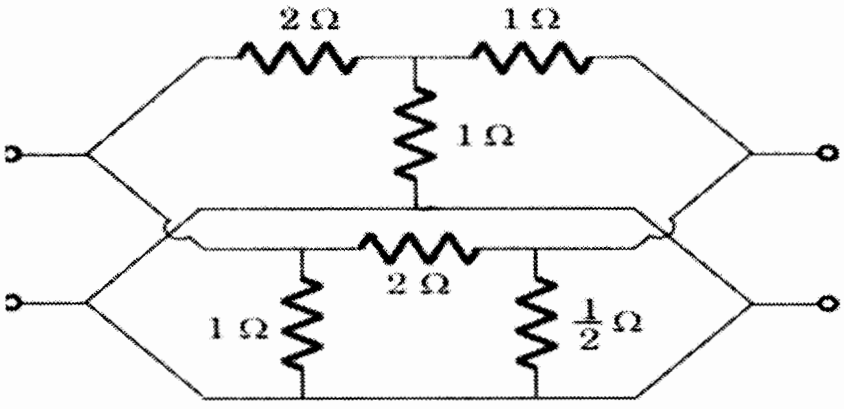
5. Attempt any *one* part of the following: (10 ×1=10)

Qno.	Question	Marks
a.	For the given circuit in fig. the value of given voltage V_O across 4*ohm* resistance. 	10

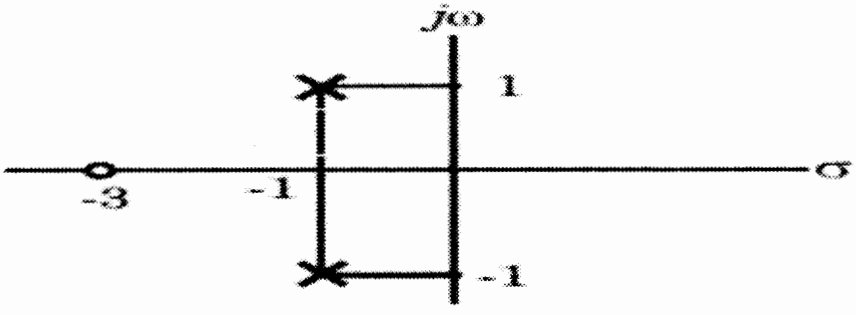
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b.	Calculate the Fourier transform of $\cos \omega_o t$ .Also Sketch its spectrum.	10

6. Attempt any *one* part of the following: (10 ×1=10)

Qno.	Question	Marks
a.	Calculate the Short Circuit Admittance Parameter of the given circuit in fig 	10
b.	Prove that for a symmetric network $Z_{11}=Z_{22}$, where Z_{11} & $Z_{22}$are Z-parameters.	10

7. Attempt any *one* part of the following: (10 ×1=10)

Qno.	Question	Marks
a.	Calculate the impedance Z(s),if Driving point impedance Z(s),of a network has pole-zero location as shown in fig . Also Z(0)=3 	10
b.	A Practical DC Current source provides 20kW to a 50 load & 20kW to a 200 load. Calculate the maximum power that can draw from it.	10

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